

RECURSION

COMPUTER SCIENCE MENTORS 61A

September 23 – September 26, 2025

1 Recursion

1. Complete the definition for `num_digits`, which takes in a number n and returns the number of digits it has. Implement the function recursively.

```
def num_digits(n):  
    """Takes in an positive integer and returns the number of  
    digits.  
  
    >>> num_digits(0)  
    1  
    >>> num_digits(1)  
    1  
    >>> num_digits(7)  
    1  
    >>> num_digits(1093)  
    4  
    """
```

2. Draw the environment diagram that results from running the code.

```
def bar(f, x):  
    if x == 1:  
        return f(x)  
    else:  
        return f(x) + bar(f, x - 1)  
  
f = 4  
bar(lambda x: x + f, 2)
```

3. Fill in `collapse`, which takes in a non-negative integer `n` and returns the number resulting from removing consecutive repeating digits, i.e. while the number may have digits repeating elsewhere in the number, the same number cannot appear next to itself.

```
def collapse(n) :  
    """  
    >>> collapse(9)  
    9  
    >>> collapse(12)  
    12  
    >>> collapse(22)  
    2  
    >>> collapse(12234441)  
    12341  
    >>> collapse(11200000013333)  
    12013  
    """  
    rest, last = n // 10, n % 10  
  
    if _____ :  
        _____  
  
    elif _____ :  
        _____  
  
    else :  
        _____
```

2 Tree Recursion

1. The *Gibonacci sequence* is a recursively defined sequence of integers; we denote the n th Gibonacci number g_n . The first three terms of the sequence are $g_0 = 0, g_1 = 1, g_2 = 2$. For $n \geq 3$, g_n is defined as the sum of the previous three terms in the sequence.

Complete the function `gib`, which takes in an integer `n` and returns the n th Gibonacci number, g_n . Also, identify the three parts of recursive function design as they are used in your solution.

```
def gib(n):  
    """  
    >>> gib(0)  
    0  
    >>> gib(1)  
    1  
    >>> gib(2)  
    2  
    >>> gib(3) # gib(2) + gib(1) + gib(0) = 3  
    3  
    >>> gib(4) # gib(3) + gib(2) + gib(1) = 6  
    6  
    """  
    if _____:  
        return _____  
    return _____
```

2. Including the original call, how many calls are made to `gib` when you evaluate `gib(5)`?

3. Gabe's Donut shop has an unlimited supply of f different flavors of donuts. Adit wants to buy a box containing d donuts. Complete the skeleton for the function `donut`, which determines the number of possible ways there are for Adit to select his d donuts from the f flavors. You may assume that d and f are non-negative integers.

Hint: Does order matter?

```
def donut(d, f):  
    """  
    >>> donut(12, 1)  
    1  
    >>> donut(12, 2)  
    13  
    >>> donut(12, 12)  
    1352078  
    >>> donut(0, 0)  
    1  
    """  
    if _____:  
        return _____  
  
    if _____:  
        return _____  
  
    return _____
```

4. Imagine you are taking the 61A midterm. There are n questions left on the exam and t time remaining. Completing question i takes i time but gives you $i + 50$ points on the exam. Write a program that will output the highest possible score on the exam given n and t (note: you can't take more than t time).

```
def midterm(n, t):  
    """  
    >>> midterm(500, 0) # No time left!  
    0  
    >>> midterm(3, 3) # 51 + 52, questions 1 & 2  
    103  
    >>> midterm(3, 5) # 52 + 53, questions 2 & 3  
    105  
    >>> midterm(4, 9) # 52 + 53 + 54, questions 2 & 3 & 4  
    159  
    """  
    if _____:  
        return _____  
    else:  
        return _____
```